Estimating peer effects in noisy, low-rank networks via network smoothing

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Oregon State University

This is joint work with my advisor



Keith Levin

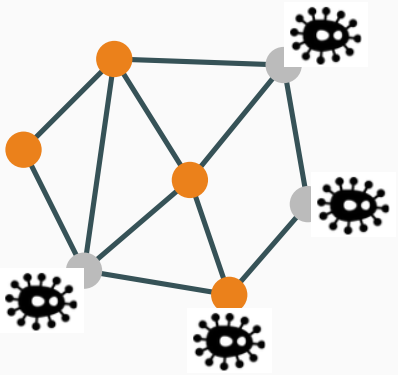
University of Wisconsin-Madison

Roadmap for today

1. What are peer effects?
2. How do we typically estimate peer effects?
3. Estimating peer effects from noisy data
4. Other current projects + pitch to students and collaborators

What are peer effects and how do we estimate them?

Peer effects are measures of social influence

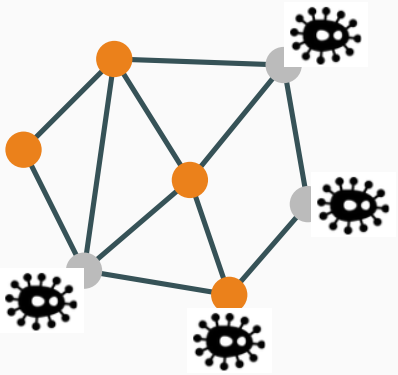


Direct effect: if I get vaccinated, I am less likely to get sick

Contagion: if my friends get sick, I am more likely to get sick

Interference: if my friends get vaccinated, I am less likely to get sick

Peer effects are measures of social influence



Direct effect: if I get vaccinated, I am less likely to get sick

Contagion: if my friends get sick, I am more likely to get sick

Interference: if my friends get vaccinated, I am less likely to get sick

* Can be defined counterfactually (Vazquez-Bare (2023)), but we do not consider counterfactual inference in this talk.

Linear-in-means models estimate social influence

Outcome (sick?) $Y_i \in \{0, 1\}$ Base rate $\alpha \in R$

$$\underbrace{Y_i}_{\text{sick?}} = \alpha$$

Linear-in-means models estimate social influence

Outcome	(sick?)	Y_i	$\in \{0, 1\}$	Base rate	$\alpha \in R$
Node degree	(num friends)	d_i	$\in \{0, 1, 2, ...\}$	Contagion	$\beta \in (-1, 1)$
Edge $i \sim j$	(friends?)	A_{ij}	$\in \{0, 1\}$		

$$\underbrace{Y_i}_{\text{sick?}} = \alpha + \beta \underbrace{\frac{1}{d_i} \sum_{j:A_{ij}=1} Y_j}_{\text{fraction sick friends}}$$

Linear-in-means models estimate social influence

Outcome	(sick?)	Y_i	$\in \{0,1\}$	Base rate	α	$\in R$
Node degree	(num friends)	d_i	$\in \{0,1,2,...\}$	Contagion	β	$\in (-1,1)$
Edge $i \sim j$	(friends?)	A_{ij}	$\in \{0,1\}$	Direct effect	γ	$\in R$
Treatment	(vaccinated?)	T_i	$\in \{0,1\}$			

$$\underbrace{Y_i}_{\text{sick?}} = \alpha + \underbrace{\beta \frac{1}{d_i} \sum_{j:A_{ij}=1} Y_j}_{\text{fraction sick friends}} + \gamma \underbrace{T_i}_{\text{vaccinated?}}$$

Linear-in-means models estimate social influence

Outcome	(sick?)	Y_i	$\in \{0,1\}$	Base rate	α	$\in R$
Node degree	(num friends)	d_i	$\in \{0,1,2,...\}$	Contagion	β	$\in (-1,1)$
Edge $i \sim j$	(friends?)	A_{ij}	$\in \{0,1\}$	Direct effect	γ	$\in R$
Treatment	(vaccinated?)	T_i	$\in \{0,1\}$	Interference	δ	$\in R$

$$\underbrace{Y_i}_{\text{sick?}} = \alpha + \underbrace{\beta \frac{1}{d_i} \sum_{j:A_{ij}=1} Y_j}_{\text{fraction sick friends}} + \gamma \underbrace{T_i}_{\text{vaccinated?}} + \underbrace{\delta \frac{1}{d_i} \sum_{j:A_{ij}=1} T_j}_{\text{fraction vaccinated friends}}$$

Linear-in-means models estimate social influence

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Linear-in-means models estimate social influence

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Letting $G = D^{-1}A$ be the row-normalized adjacency matrix, we can write this in matrix-vector form:

$$Y = \alpha \mathbf{1}_n + \beta GY + T\gamma + GT\delta + \epsilon$$

The canonical two-stage least squares linear-in-means estimator

$$\hat{\beta}_{\text{TSLS}} = (Q^\top M Q)^{-1} Q^\top M Y$$

where

$$Q = [\mathbf{1}_n \; W \; GW \; GY] \qquad\qquad G = D^{-1} A$$

is the covariate matrix and

$$M = H(H^\top H)^{-1} H^\top$$

is the projection onto the span of the instrument matrix

$$H = [W \; GW \; G^2 W \; G^3 W].$$

The problem

This talk is about the linear-in-means model

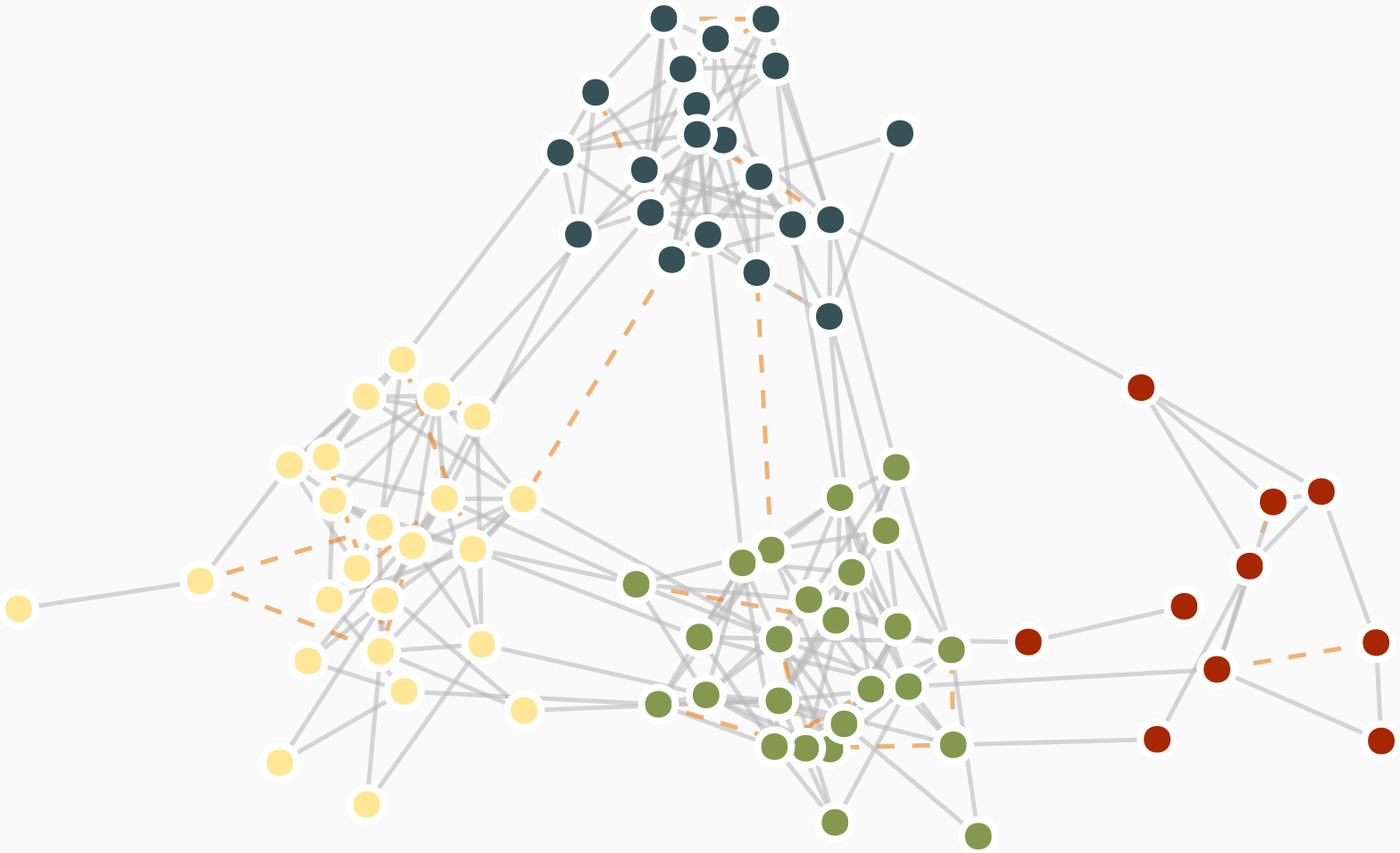
Very popular tool for estimating social influence

Used in education, crime, health, social policy, etc.

I discovered an issue with this model

E.g., (Sacerdote, 2001), (Epple & Romano, 2011), (Soetevent & Kooreman, 2007), (Trogdon et al., 2008), (Duflo & Saez, 2003), (Bertrand et al., 2000), (Glaeser et al., 1996), (Patacchini & Zenou, 2012), (Carrell et al., 2013), etc.

Peer effects in networks with missing and noisy edges



A new estimator for the linear-in-means model

Setting: Treatment random and independent of network. $T_i \overset{\text{iid}}{\sim} \text{Bern}(0.5)$

$$Y = \alpha \mathbf{1}_n + \beta GY + T\gamma + GT\delta + \epsilon$$

We started to run a simulation study to confirm that our estimator worked…

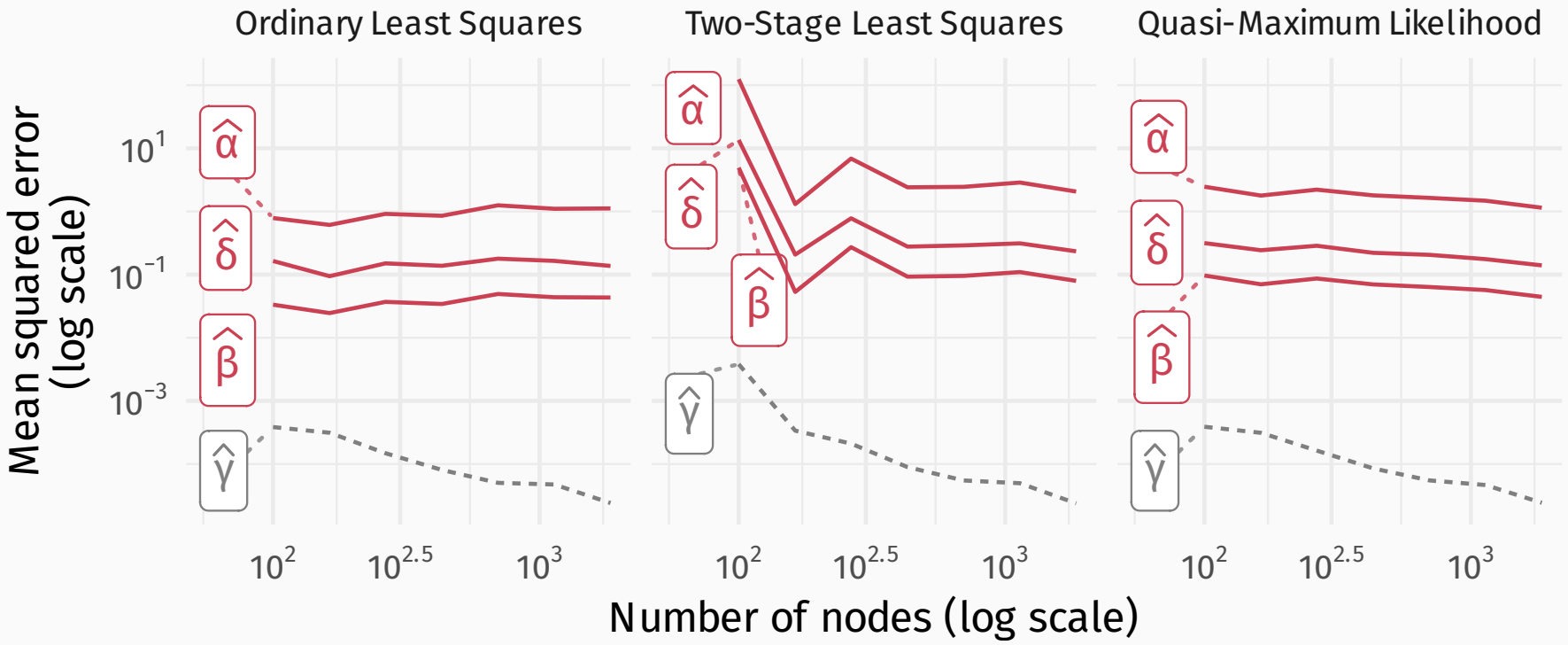
Generate Y via the reduced-form specification $Y = (I - \beta G)^{-1}(\alpha \mathbf{1}_n + \gamma T + \delta GT + \epsilon)$.

Base rates and interference are collinear in large samples

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \underbrace{\begin{bmatrix} \mathbf{1}_n & GY & T & GT \\ 1 & GY_1 & 1 & 1/2 \\ 1 & GY_2 & 0 & 1/2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & GY_n & 1 & 1/2 \end{bmatrix}}_{\text{as } n \to \infty} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

Sometimes can't distinguish between base rate α and interference δ

In simulations, we couldn't estimate β either



Causal folks study treatments that depend on network position

Influence

It is an ancient observation that people are influenced by others (nearby) in their social network—that is, the behavior of one node in a social network adapts or responds to that of neighboring nodes. Such social influence is not just a curiosity, but of deep theoretical and empirical importance across the social sciences. It is also of great importance to various kinds of social engineering, for example, marketing (especially, but not only, "viral" marketing), public health (over-coming "peer pressure" to engage in risky behaviors, or using it to spread healthy ones), education ("peer effects" on learning), politics ("peer effects" on voting), etc. Conversely, it is an equally

bias.[1]

Shalizi and Thomas (2011, sec. 4.3) did hint at a possible approach for identification of social influence, even in an homophilous network. When a network forms by homophily, a node is likely to be similar to its neighbors. Following this logic, these neighbors are likely to be similar to *their* neighbors and therefore the original node. In the simplest situations, where there are only a limited number of node types, this means that a homophilous network should tend to exhibit clusters with a high within-cluster tie density and a low density of ties across clusters. Breaking the network into such clusters might, then, provide an observable proxy for the latent homophilous attributes. The same idea would work, *mutatis mutandis*, when

Dependence between treatment and network might help

$$\underbrace{[GT]_i}_{\text{fraction vaccinated friends}} = \underbrace{\frac{1}{d_i} \sum_{j:A_{ij}=1} T_j}_{\text{average of dependent treatments}}$$

GT might not converge, or might converge to a non-constant value

Stochastic blockmodels induce dependence



Stochastic blockmodels are an intuitive way to induce dependence

Block indicators Z_i

Popularity parameters ω_i

Mixing matrix $B \in [0,1]^{r \times r}$

$$\mathbb{P}(A_{ij} = 1 \mid Z, \omega) = \omega_i Z_i B Z_j^\top \omega_j$$

More takeaways

Takeaway 3

Explicitly modelling dependence between nodal covariates and network structure can aid identifiability and resolve asymptotic collinearity issues.

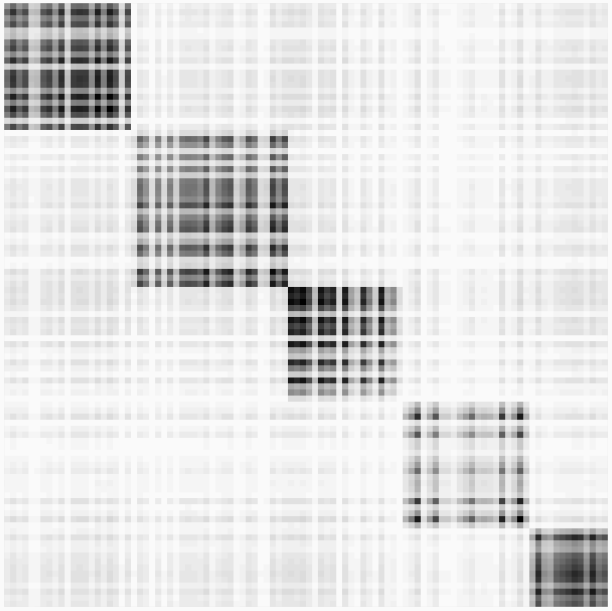
Takeaway 4

Treatments dependent on network must be considered on a case-by-case basis, considering both the treatment and the network model.

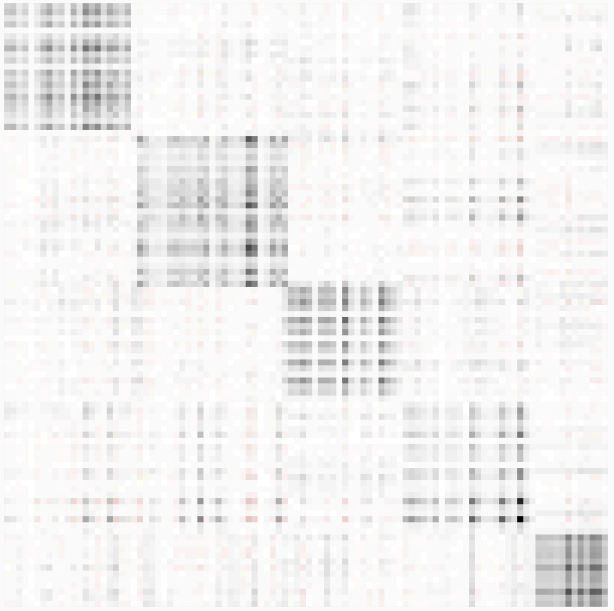
Peer effects in networks with missing and noisy edges



A

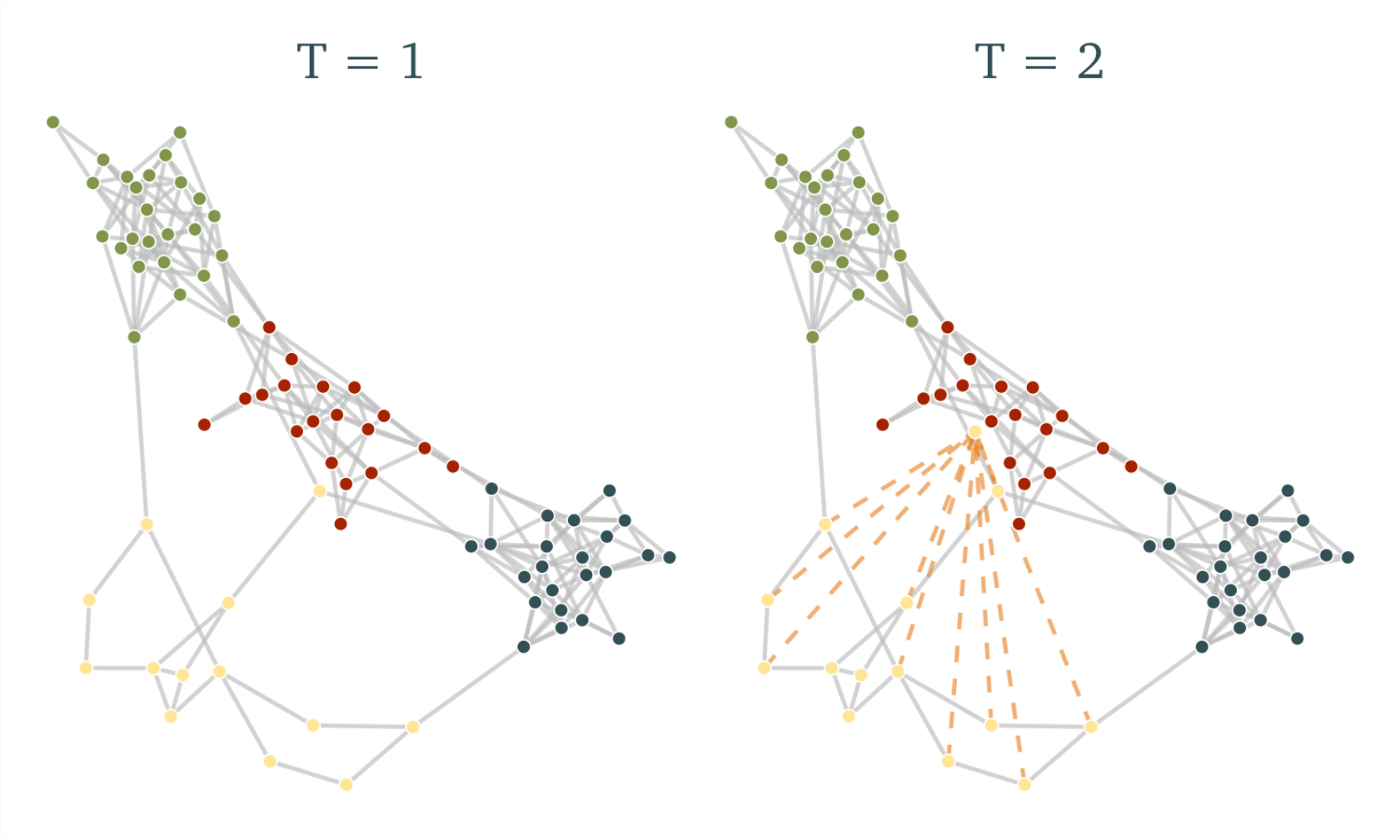


$\mathbb{E}[A \mid X] = XX^\top$



$\hat{X}\hat{X}^\top$

Causal inference for co-evolution models



Estimating peer effects from noisy data

Humans constantly influence their social contacts



…but measuring someone's friends is challenging

If I ask you who your friends are:

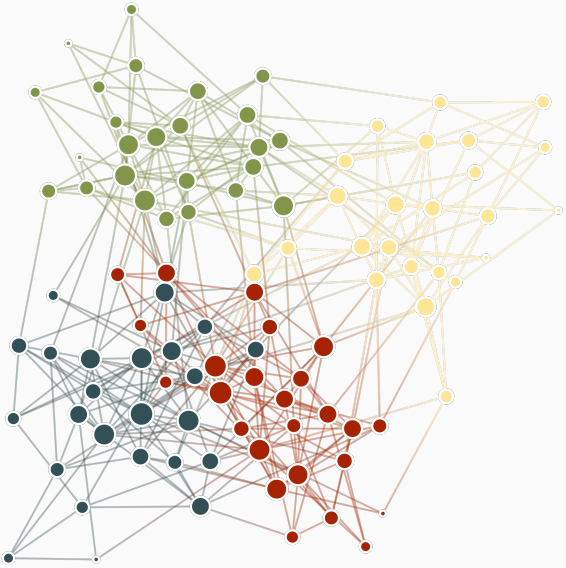
- answers vary depending on how exactly I ask you,
- you'll probably forget day-to-day social contacts, and
- you might think someone is your friend, but they might disagree.

We use estimated **likelihood of friendship** instead

$$Y_i = \beta_0 + W_i\beta_{\mathrm{w}} + X_i\beta_{\mathrm{x}} + \underbrace{\beta_{\mathrm{y}} \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{peer contagion}} + \epsilon_i \qquad \text{requires all } A_{ij}$$

$$Y_i = \theta_0 + \underbrace{W_i\theta_{\mathrm{w}}}_{\text{observed covariate effects}} + \underbrace{X_i\theta_{\mathrm{x}}}_{\text{latent positional effect}} + \underbrace{\theta_{\mathrm{y}} \sum_{j \neq i} \frac{X_i^\top X_j}{\mathbb{E}[d_i \mid X]} Y_j}_{\text{latent contagion}} + \epsilon_i \qquad \text{requires } \mathbb{P}(A_{ij} = 1)$$

…based on a low-rank network model



Intuition: stochastic blockmodels

Block indicators Z_i

Popularity parameters ω_i

Mixing matrix $B \in [0,1]^{r \times r}$

$$\mathbb{P}\big(A_{ij} = 1 \mid Z, \omega, B\big) = \underbrace{\omega_i Z_i B Z_j^\top \omega_j}_{\text{rank } r}$$

Only assume undirected $A \in R^{n \times n}$ with low rank $P = \mathbb{E}[A \mid X] = XX^\top$ and sub-gamma (Bernoulli, Poisson, Gaussian, Gamma, …) deviations around A.

Estimating contagion with noisy data

1. Humans constantly **influence each other**. We want to **estimate how strong social influence** is.
2. However, **measuring friendships is challenging**.
3. To estimate social influence, we **need to know who is friends with whom**.
4. We propose **a method that works with noisy friendship measurements**.

…to estimate how much people influence each other

Let $A \approx \hat{U}\hat{S}\hat{U}^\top$ be the rank r SVD of A. Put $\hat{X} = \hat{U}\hat{S}^{1/2}$ and $\hat{P} = \hat{X}\hat{X}^\top$ and $\hat{d}_i = \sum_j \hat{P}_{ij}$ and $\hat{D} = \text{diag}\left(\hat{d}_1, ..., \hat{d}_n\right)$ and $\hat{G} = \hat{D}^{-1}\hat{P} \in R^{n \times n}$. Let $\hat{Q} = \left[W \;\; \hat{X} \;\; \hat{G}Y\right]$ and $\hat{H} = \left[W \;\; \hat{X} \;\; \hat{G}W \;\; \hat{G}\hat{X} \;\; \hat{G}^2W \;\; \hat{G}^2\hat{X}\right]$. Let $\widehat{M} = \hat{H}\left(\hat{H}^\top\hat{H}\right)^{-1}\hat{H}^\top$. Then

$$\hat{\theta}_{\text{TSLS}} = \left(\hat{Q}^\top\widehat{M}\hat{Q}\right)^{-1}\hat{Q}^\top\widehat{M}Y$$

Theorem: $\hat{\theta}_{\text{TSLS}}$ is asymptotically normal under the latent contagion model

$$\sqrt{n}\left(\hat{\theta}_{\text{TSLS}} - \boldsymbol{\theta}\right) \to \mathcal{N}(0, \Sigma)$$

This can be as good as knowing exact friendship!

Theorem: $\hat{\theta}_{\text{TSLS}}$ is asymptotically normal under the peer contagion model

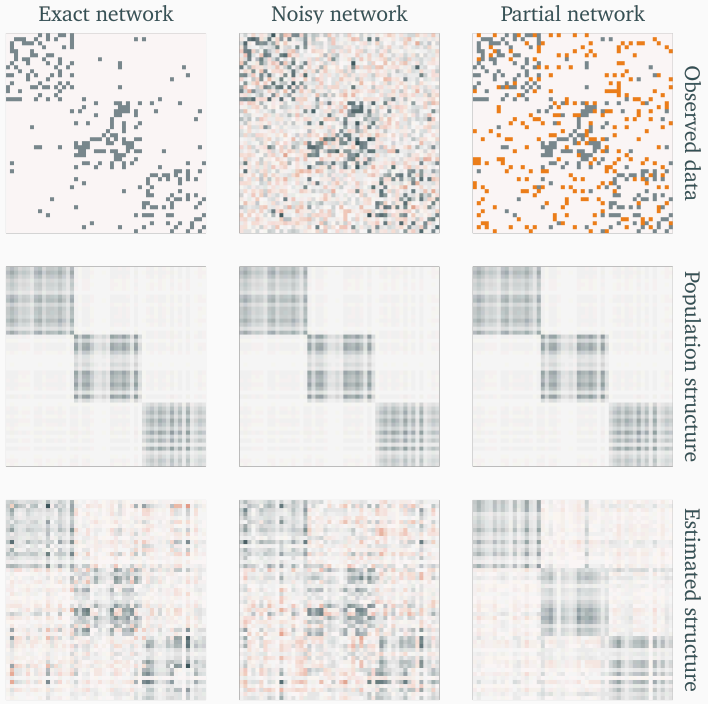
$$\sqrt{n}\left(\hat{\theta}_{\text{TSLS}} - \beta\right) \to \mathcal{N}(0, \Sigma)$$

How generally can we apply the strategy of replacing A with $\mathbb{P}(A)$?

More about Alex Hayes

- I'm an Assistant Professor in Statistics & Data Science
- I develop statistical methods for social networks and causal inference
- I write lots of R code and worked on some of the tidyverse packages
- I'm actively looking for collaborators!

Our estimators can handle noisy measurements



…because latent structure is estimable at fast rates even in noisy networks

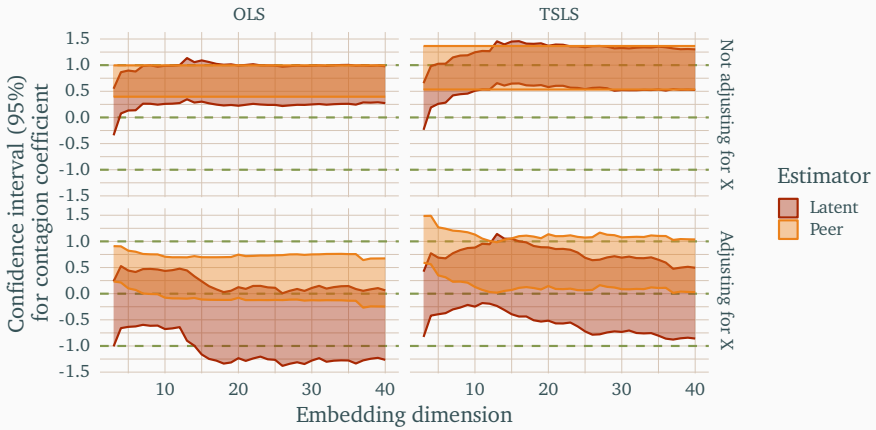
Anxiety before final exams amongst MBA students

Data on 241 MBA students from "Do Your Friends Stress You out? A Field Study of the Spread of Stress…" by Shihan et al (2023)

$$\text{anxiety} \sim \text{personality} + \text{demographics} + X + \text{latent peer anxiety}$$

$$\text{anxiety} \sim \text{personality} + \text{demographics} + X + \text{peer anxiety}$$

X accounts for localized/group effects. Dimension of X is unknown!



Model specification dramatically impacts inference

My background and research

My research interests and professional experience

Research interests:

- (Social) networks
- Causal inference
- Peer effects
- Social processes in science
- Software design

Professional background:

- PhD Statistics, UW-Madison
- Facebook Core Data Science (Intern)
- RStudio (Intern)
- B.A. Statistics, Rice University

Thank you! Questions? Comments?

Network data is expensive, ARD is cheap

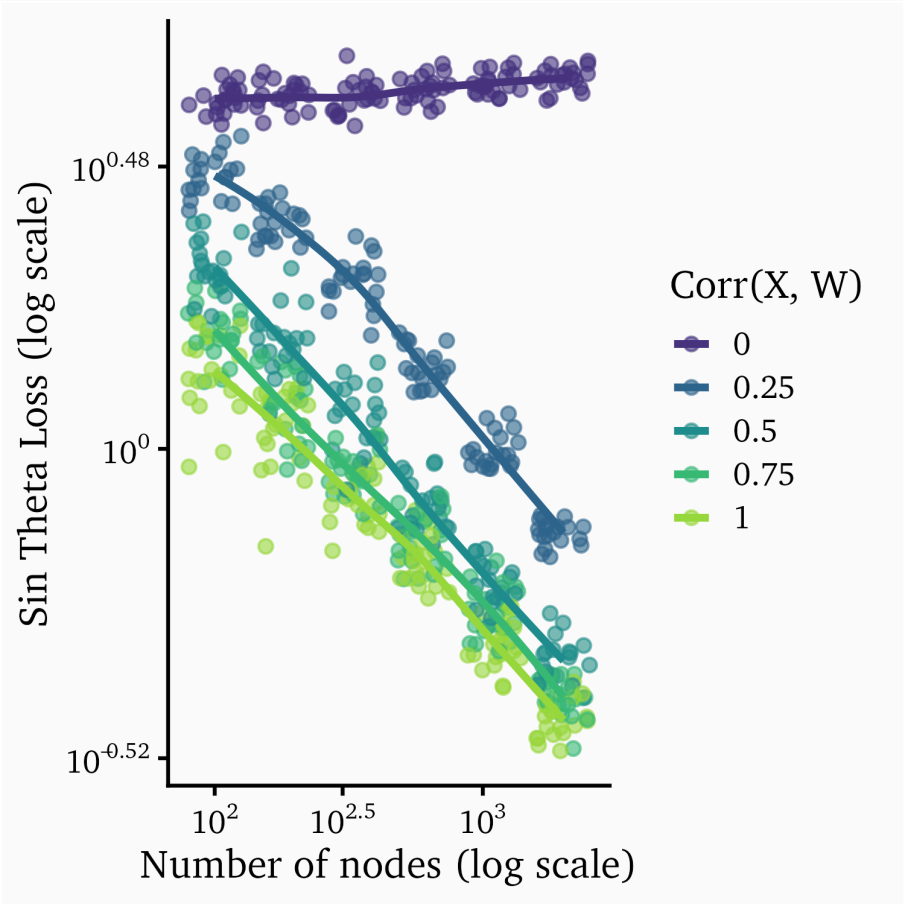
Two stage spectral estimation is convenient and enables full uncertainty quantification in downstream analysis

Good traits are strongly homophilous

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github.com/alexpghayes

Appendix

Assumptions of the canonical two-stage least squares estimator

1. All diagonal elements of G are zero
2. $(I - \beta G)^{-1}$ is non-singular with $|\beta| < 1$.
3. G and $I - \beta G$ are uniformly bounded in row and column sums.
4. The regressor matrices W have full column rank (for n large enough), and the elements of W are uniformly bounded in absolute value.
5. ϵ_i are i.i.d. with zero mean and bounded variance $\mathbb{E}[\epsilon_i^2] = \sigma_\epsilon^2 < b < \infty$. Additionally ϵ_i have finite fourth order moments.

Assumptions of the canonical two-stage least squares estimator

Further, the instruments H have full column rank and are composed of a subset of linearly independent columns of $(W \;\; GW \;\; G^2W)$ where the subset contains W and X. Additionally

$$\lim_{n\to\infty} \frac{1}{n} H^\top H$$

is finite and non-singular and

$$\frac{1}{n} H^\top Q$$

converges in probability to a matrix that is finite and has full column rank.

Assumptions of the canonical two-stage least squares estimator

If all the previous assumptions hold, then

$$\sqrt{n}\left(\hat{\beta}_{\text{TSLS}} - \beta\right) \to \mathcal{N}\left(0, \sigma_\epsilon^2 (Q^\top M Q)^{-1}\right)$$

recalling that

$$Q = [\mathbf{1}_n \; W \; GW \; GY] \qquad\qquad G = D^{-1}A$$

is the covariate matrix and

$$M = H(H^\top H)^{-1} H^\top$$

is the projection onto the instruments $H = [W \; GW \; G^2W \; G^3W]$.

Estimators

- OLS: lm(y ~ Gy + T + GT)
- TSLS: ivreg(y ~ Gy + T + GT | T + GT + G2T), with instruments T, GT, G^2T

Random dot product graphs

Definition (Young & Scheinerman (2007)). Let F be a distribution on R^r such that $0 \leq x^\top y$ for all $x, y \in \text{supp } F$ and the convex cone of supp F is r-dimensional. Draw $X_1, X_2, ..., X_n \overset{\text{iid}}{\sim} F$, and collect these in the rows of $X \in R^{n \times r}$ for ease of notation. Conditional on these n vectors, which we call latent positions, generate edges by drawing $\{A_{ij} : 1 \leq i < j \leq n\}$ as independent (ν, b)-subgamma random variables with $\mathbb{E}\left[A_{ij} \mid X\right] = \rho X_i^\top X_j$, where $\rho \in [0, 1]$. Then we say that A is distributed according to an n-vertex random dot product graph with latent position distribution F, (ν, b)-subgamma edges and sparsity factor ρ. We write $(A, X) \sim \text{RDPG}(F, n)$.

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